

Figure 1

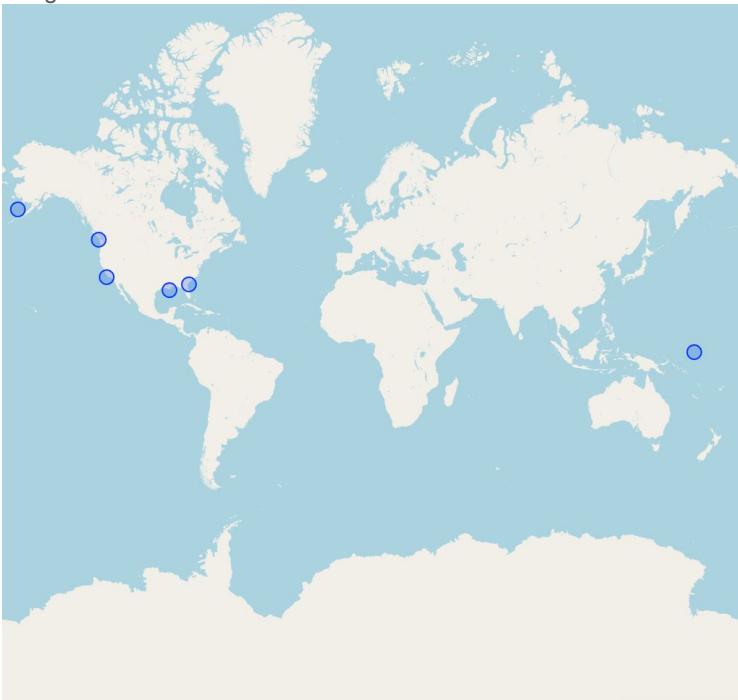


Figure 2

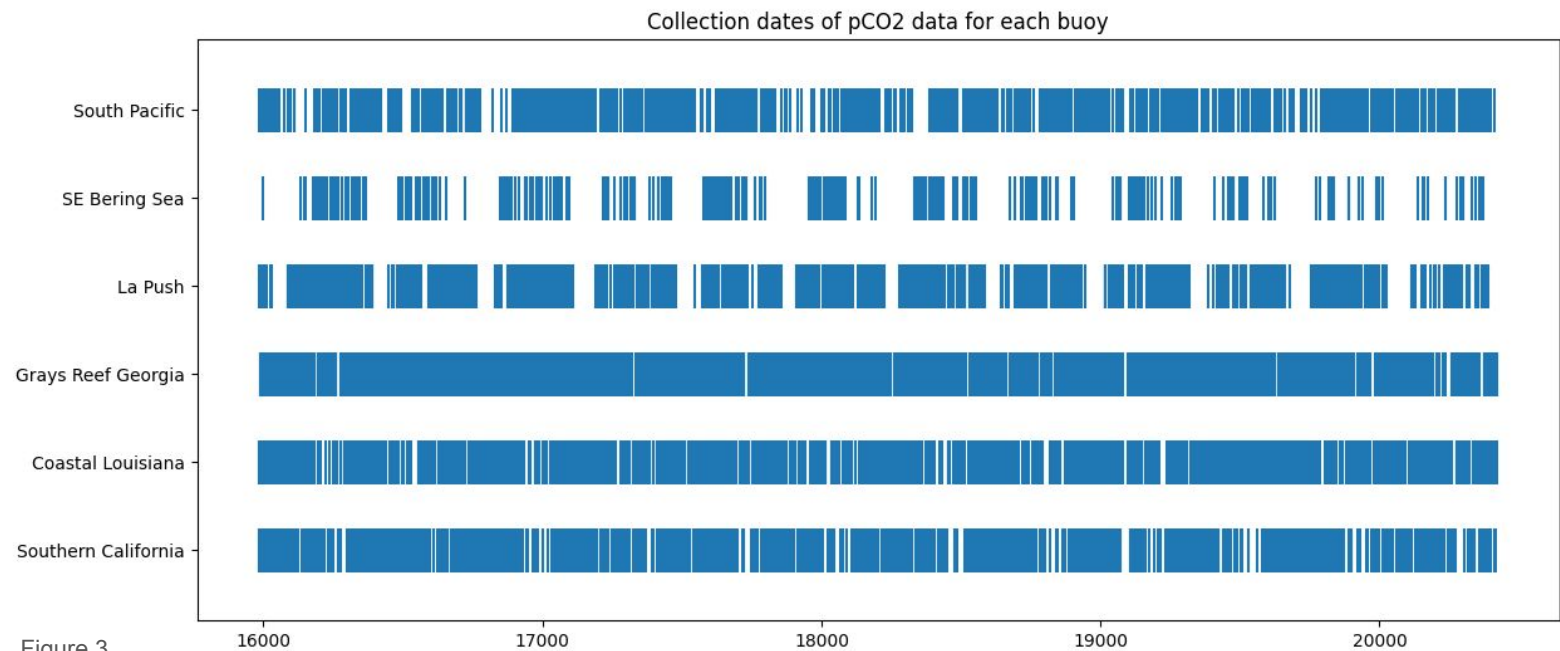
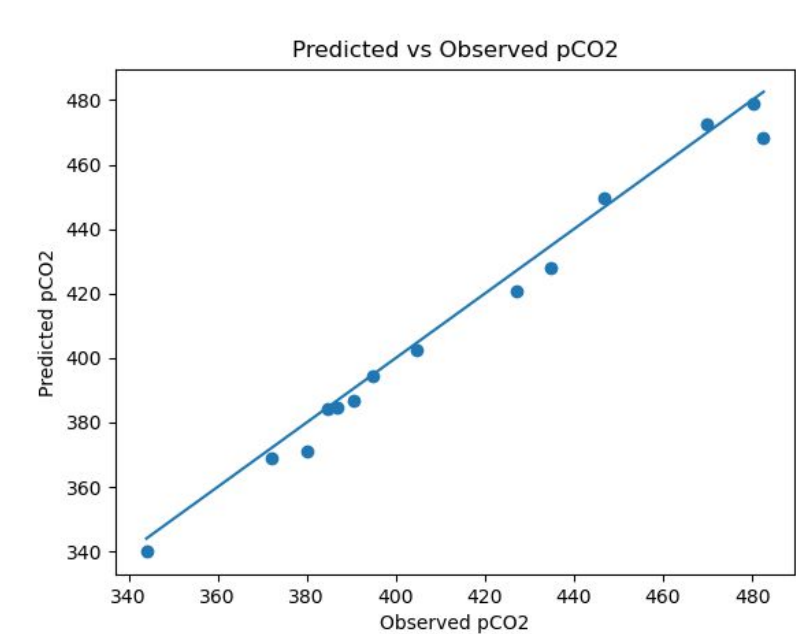


Figure 3



Citations:
Microsoft. (2026). Conversation with AI assistant about data visualization techniques in Python [Generative AI chat]. Copilot. <https://copilot.microsoft.com>
M. Ranu, Geocanographers, (2026), GitHub repository, <https://github.com/cbw03/Geocanographers>

- Figure 1: Map showcasing locations of all 6 buoys surveyed for this project. Spatial distribution affects sea surface temperature throughout the year, altering the chlorophyll-a content on the ocean's surface, which is a marker for pCO₂. Due to availability in buoys, the model is trained on data bordering the United States as well as one buoy east of Papua New Guinea.
- Figure 2: An event plot showcasing the distribution of pCO₂ collection dates across all of the buoys used as verification data within our model. Each buoy varied greatly in their data consistency, leading to less verification in areas surrounding SE Bering Sea (northern-most buoy) especially. (Figures 1&2 code was edited by the built in Copilot application within VSCode)
- Figure 3: A plot showing how close the model came to a 1:1 pCO₂ output vs the recorded pCO₂ at our buoy locations. The predicted values are consistently lower than the observed values from the buoys; other parameters that may bring these values closer are ocean current measurements and salinity, which both have a relationship to how much CO₂ can/will be contained in water. (sourced from M.Ranu)

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M. Ranu, Geoceanographers, (2026), GitHub repository, <https://github.com/cbw03/Geoceanographers>